# Degly Sebastián Pava Pava

deglypavapava@gmail.com https://deglypava.com IMLEX and Systems and Computer Engineer

From: August 2020
To: September 2023
(Currently in Japan)

## Erasmus Mundus Japan – Master of Science in Imaging and Light in Extended Reality (IMLEX)

- *Quality Board Member and Representative*
- *Triple Degree Program (Msc. Optics, Image, Computer Vision, Msc. Engineering, Msc. Computer Science)*
- *Toyohashi University of Technology (Japan), KU Leuven (Belgium), University of Eastern Finland (Finland), University Jean Monnet Saint-Etienne (France)*

***Photonics | Realtime 3D Rendering | Computational Imaging | CUDA | OpenMP | Computer Graphics | Physical Optics | Machine Learning | ARKit | ARCore | VR/AR.***

## WORK EXPERIENCE

Company Owner

### Omni Applications L.L.C. (Florida, U.S.A) – Lucid Dreams Ltd. (Bogotá, Colombia)

- ***Tech: ThreeJS | Unity | Unreal | SparkAR |360 Videos | Insta360Studio | LensStudio | Blender | Jira | Slack***

From: January 2018
To: August 2020

### Lead XR Developer | Hackspace for Innovation and Visualization in Education

*Faculty of Medicine, The University of British Columbia, Vancouver, B.C. (Canada)*

- As part of THE HIVE Research Laboratory, contributed to The HoloBrain and MR Manikin / OCS projects by creating multiple 3D experiences and integrating them into a mixed reality environment using concepts of 3D visual computing. Utilized Unity 3D Engine to develop the experiences and released them through various devices, including Oculus Rift, HTC VIVE, and Extended Reality devices.
- ***Tech: Unity | Microsoft HoloLens 1&2 | OpenXR | Mixed Reality Toolkit (MRTK) | Vuforia | Blender | Neuroanatomy | Azure | Cloud***

From: February 2017
To: September 2019

### Lead Developer and Engineer at Pontifical Xavierian University

*Department of Industrial Engineering and Systems Engineering at Pontifical Xavierian University, Bogota (Colombia)*

- Developed a Material Requirements Planning (MRP) simulation system using NetLogo programming language, which effectively manages production planning, scheduling, and inventory control processes in a 3D environment. The MRP simulation is currently utilized in production classes to train new students in the concepts of MRP and scheduling.
- ***Tech: NetLogo | Anylogic | Linear Algebra | Microsoft Office Suite | Adobe Suite | Research Techniques***

## EDUCATION AND TRAINING

**Master's studies:**

*IMLEX Award + MANUTECH Sleight Scholarship + ERASMUS Mundus Japan Scholarship*

**Bachelor studies:**

*Systems and Computer Engineer. Special Admission Program, best high school graduates of Colombia Scholarship at Universidad Nacional de Colombia.*

**Trained in multiple research groups:**

### HIVE Research Lab – ELAP (Emerging Leaders of the Americas Program) Scholarship

*The University of British Columbia, Vancouver, B.C. (Canada)*

From: January 2019
To: April 2019

- Emerging Leaders of the Americas Program Scholarship winner offered by the Canadian Government.
- Visiting International Research Scholar Fellowship.
- Developed neuroanatomy-based applications while making use of MR and VR technologies for research and educational purposes.

From: June 2018
To: December 2018

### Gold IronHacks Purdue-UNAL 2018 Fellowship – RCODI

*Purdue University, West Lafayette, Indiana (United States)*

- Participated on the team that organized the Gold IronHacks Purdue-UNAL Hackathon 2018.
- Created and managed parts of the research such as surveys and privacy agreements.
- Part of the judges that would choose the best implementation of Open Data within a participant created website. Using a defined set of attributes, we analysed how the data was implemented by students.

| Digital skills | Extended Realities | Programming | Optics / Physics | Research and Tools |
|---|---|---|---|---|
| | Unity \| Blender \| ThreeJS | C# \| Python \| C++ | Photonics \| Colorimetry | ImageJ3D \| Healthcare IT |
| | AWS \| Android \| UX design | JavaScript \| NetLogo | Eye-Tracking \| Robotics | TensorFlow \| Photon Realtime |
| | Vuforia \| XR SDKs \| Arduino | Git (GitHub) \| WebXR | Spectral Imaging | Mendeley \| Adobe Suite |

## PERSONAL SKILLS

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|---|---|---|---|
| **Spanish** | Native proficiency | **French** | Limited Working Proficiency |
| **English** | Bilingual - IELTS (8.0) and iBT TOEFL (103 points) | **Japanese** | Furigana / Basic Speaking and Writing |